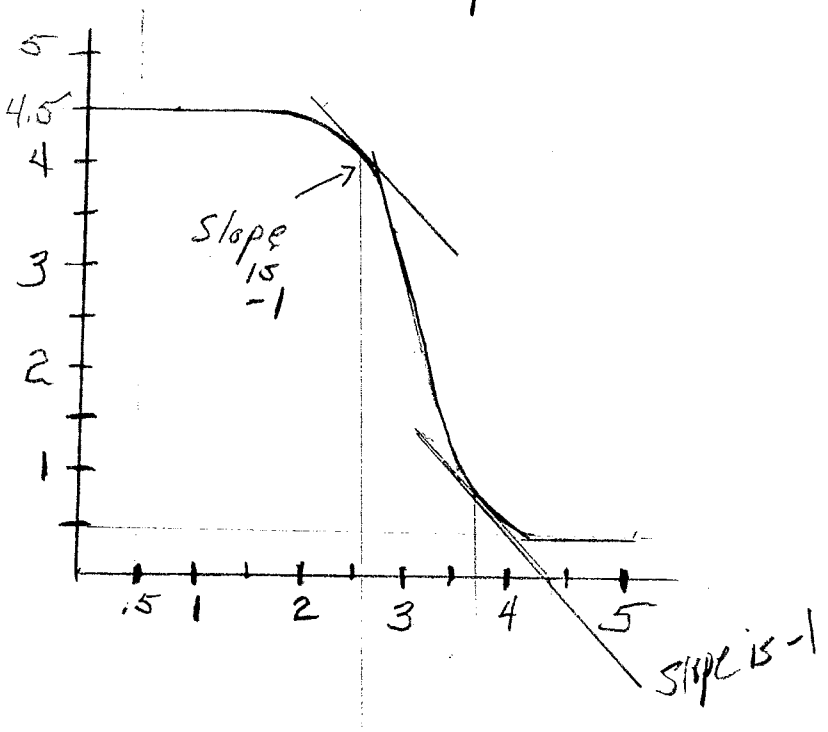
Test 1 SC571 Spring 2011 _____ **7 pages and 15 problems.**
Name Grading & Solutions

Note: a 1x inverter has a 2x2 NMOS pulling down and a 2(long) x 4 (wide) PMOS pulling up.

1. **[5] Assume Vdd=5V. Assume a fully-complementary CMOS inverter. The input/output curve is drawn below. What is the noise margin on the high and the low side?**

2.5 points each



Hi Side : 2.6 - .5 = 2.1

Low Side : 4.5 - 3.7 = 0.8

Low & Hi can be reversed

If 5: (-1)

2. **[6] What is the apparent pullup resistance of the pullup transistor and the apparent pulldown resistance of pulldown transistor if this is the output of an inverter that drives a 200 pF load?**

~1 to ~3 is OK

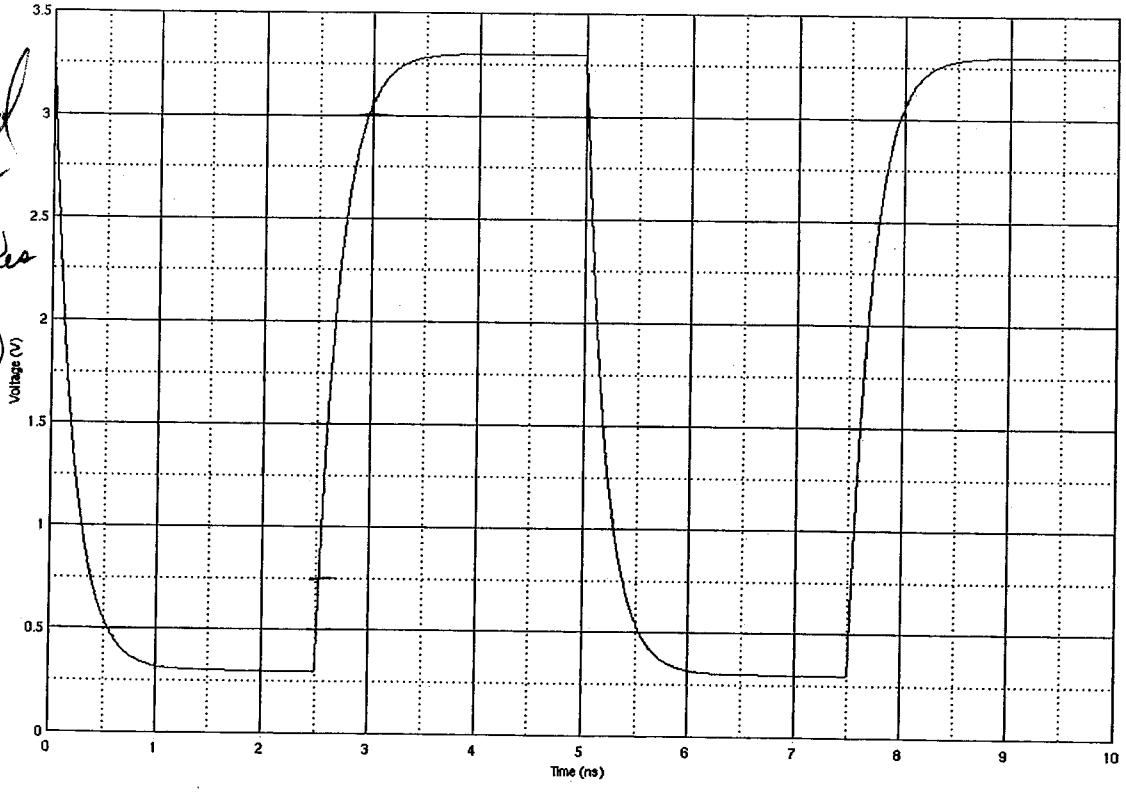
$R\,200\times10^{-12} = 0.5\times10^{-9}$

$R = \frac{5\times10^{-10}}{2\ 10^{-10}} = 2.5$

(3) **Rpullup=**_____

used ~2.5 both times

-4



(3) **Rpulldown=** Same

(2) (2)

3. **[6] In the case of two identically-sized poly and diffusion wires, which one has the greatest total capacitance? Explain your choice and in your explanation, address the issue of side-wall capacitance.**

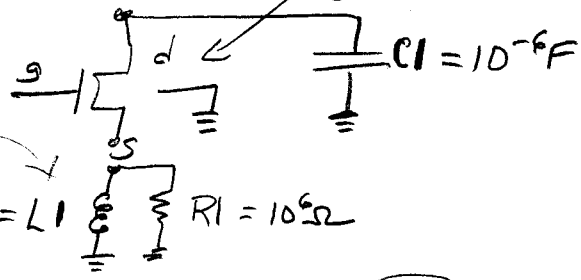
(2)

topology is worth 4. (-1)

4. **[6]SPICE Draw the circuit element that corresponds to the following line of SPICE netlist.**

M1 d g s gnd nmos Vto=1V

L1 s gnd 1e7
C1 d gnd 1e-6
R1 s gnd 1e6



-1 if source not to ground

labels (-1)

Nothing off for component names

5. **[6]Explain drain capacitance. Include a sketch and include discussion of the capacitance looking straight down and also the side wall capacitance. Include a reference to what is the mechanism involved in forming "plates" of the capacitance.**

2 1 1

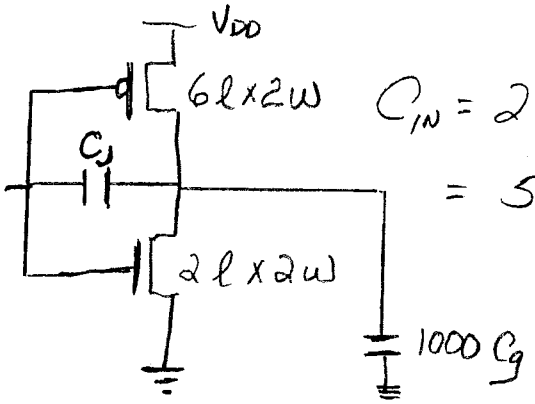
overlap SD, not Drain: (-6)

6. **[6] What is the input capacitance to the inverter sketched below? Take the Miller effect into account, utilizing *the physical circuit element Cj* which is drawn in the circuit. Cj=Cg/2.**

$C_g \cdot 5 + 1000 C_g$: (-2)

each gate that's wrong (-2)

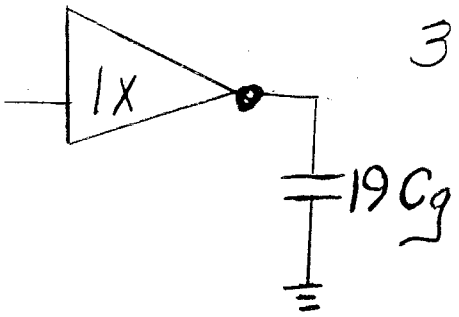
$C_{MILLER} = C_g/2$ (-1)



$C_{IN} = 2C_j + C_g + 3C_g$

$= 5C_g$

7. **[4]To a first approximation, what is the input capacitance seen looking into the inverter shown below.** [redacted]



3 Cg anything else (-4)

8. **[5]Sheet resistance. A poly wire of dimensions 2 x 800 has a sheet resistance of 10 ohms. What is its resistance? Assume the current goes through the wire via the "long" dimension.**

$R = 10 \cdot \frac{800}{2} = 4K\Omega$

arith (-1)

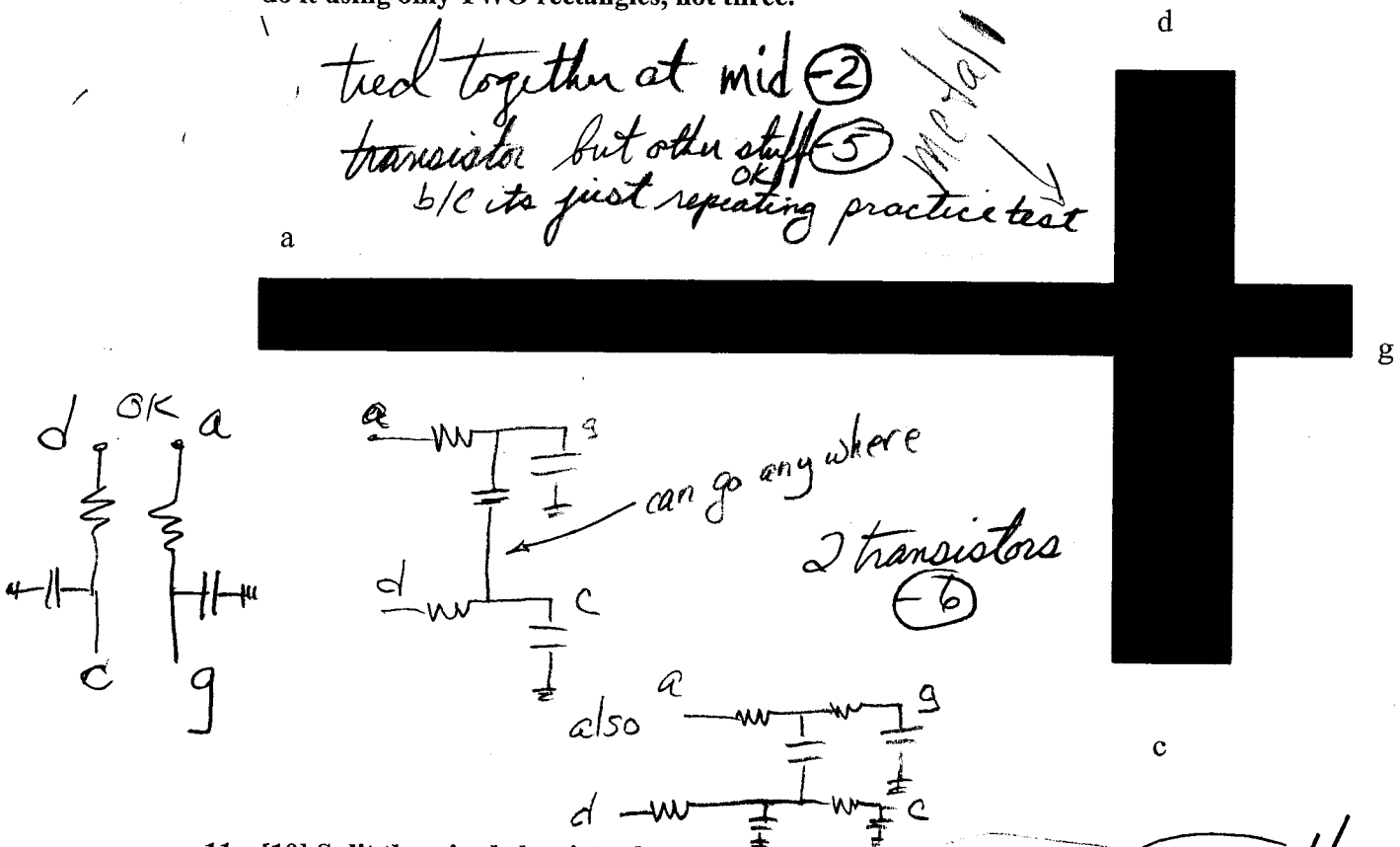
9. **[4]Capacitance of wires and gates. The poly wire above(in problem 8) has a capacitance of 1.0 fF for each 2x2 =4 square units of area. What is the total lumped capacitance of this wire. Neglect fringing capacitance and edge capacitance.**

$C = \frac{2 \times 800}{4} \times 10^{-15} F = 400 \times 10^{-15} F = 4 \times 10^{-13} F$

arith (-1)

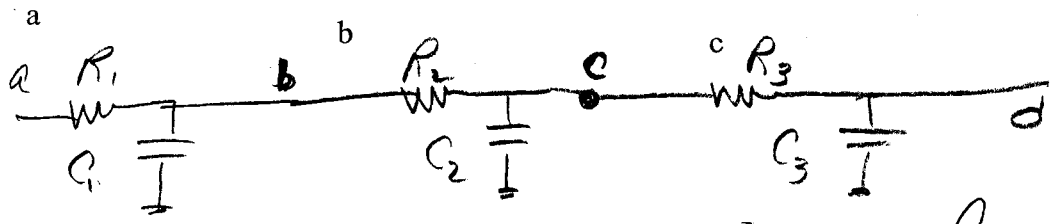
no div 4 (-2)

10. **[6]** Extract (draw) a circuit model. The nodes you will use are {a,c,d,g}. Make up your own labels for the extracted elements. If you drew this on a screen, you would do it using only TWO rectangles, not three.



11. **[10] Split the wire below into three equal lengths. First draw the extracted circuit. and then write the Elmore expression for the delay from a to c. . Do not estimate the numerical value of circuit elements. Just give them symbolic names, e.g. C1, R1, C2, R2, C3, R3.**

4



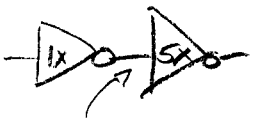
$T_{ac} = R_1[C_1 + C_2 + C_3] + R_2[C_2 + C_3]$. Since all R's & C's are the same $5R_1C_1$

accept also R_2C_2

If R_3 is included, -4

No C_3 anywhere -3

12. **[11] The tau model. A 1x fully-complementary CMOS inverter drives an identical 5x inverter. (A 5x inverter means that the widths of pull-ups and pulldown transistors are of width 10 for the NMOS transistor and 20 for the PMOS transistor. The gate length is always 2.) What is the pulldown time at the output of the first inverter? Use Tau_nmos_2= 0.1ns. Use 0.2ns for the PMOS transistor.** ***Assume (and include in your calculations) a drain loading of the 1x inverter is 1Cg.***



$T_{PD} = 0.1ns \frac{C_L}{C_D} = 0.1ns \frac{15C_g + 1C_g}{1C_g}$

$= 1.6ns$

C_L wrong (-4)

$2C_{D_{drn}}$ (-2) ↔ No C_{DRAIN}

C_{Driver} wrong (-3)

arithmetic (-1)

13. **[11] The RC model. A 1x fully-complementary CMOS inverter drives a 100fF load. Assume the NMOS transistor has a sheet resistance of 10kohm and PMOS transistors have 20kohm, what is the pulldown time constant?**

$\tau_{PD} = 10^4 \times 10^{-13} = 1ns$

arithmetic (-1)

PU: (-2)

14. **[4] What is the total dynamic power consumed by a 1x inverter that drives a 10 Cg load at a clock frequency of 10 MHz. Assume Vdd = 5V and the clock also operates between 5 V and 0 V with a fast rise and fall time, and there is time for the signals to settle to a final value following clock transitions.**

total load is 13 Cg

$CV^2f = 13C_g \cdot .25 \cdot 10^{7}\ W$

$= 13 \cdot 25 \times 10^{-7}\ W$

Missed clock load (-1)

√ Square (-1)

Arithmetic -1

15. **[5] Explain what is velocity saturation of electrons in a short-channel NMOS transistor.**

variable

16. **[5] Explain *what* (3) is drain induced barrier lowering (DIBL) and *why* (2) *is it significant* for "short-channel" transistors.**

variable

Many got hot electrons in here
Only one nailed it.